A Self-Describing Cyclic Number: The Unique Cyclotomic Decomposition of 142857 into Hamming-Family Scaffold Parameters

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June 2026

Abstract

The cyclic number $142857 = (10^6 - 1)/7$ has prime-power factorization $3^3 \cdot 11 \cdot 13 \cdot 37$. We show this equals $r^3(n+k)(2n-1)(2^k + nr)$ where $r = 3$, $n = 7$, $k = 4$ are the parameters of the $[7, 4, 3]$ binary Hamming code and $b = 10 = n + r$ is the base. We prove this cyclotomic–Hamming alignment is unique: among all binary Hamming-family scaffolds $\{2^r - 1,\ 2^r - 1 - r,\ r\}$ with the canonical base $b = n + r$, only $r = 3$ produces a cyclic number whose cyclotomic factor-blocks decompose into code parameters. The key uniqueness condition, $2^{r-1} = r + 1$, has the single positive integer solution $r = 3$. Two independent conditions from other cyclotomic factors confirm this. A computational null test through $p = 47$ verifies that no other full-reptend prime in base 10 exhibits comparable factorization alignment.

1 Introduction

The number 142857 is the repetend of 1/7 in base 10 and is well-known as a cyclic number: multiplication by 1 through 6 permutes its digits cyclically. Its prime factorization is

$$142857 = 3^3 \cdot 11 \cdot 13 \cdot 37.$$

We observe that each prime-power block admits a low-complexity expression in the parameters of the $[7, 4, 3]$ Hamming code:

$$27 = 3^3 = r^3, \tag{1}$$
$$11 = 7 + 4 = n + k, \tag{2}$$
$$13 = 2 \cdot 7 - 1 = 2n - 1, \tag{3}$$
$$37 = 2^4 + 7 \cdot 3 = 2^k + nr. \tag{4}$$

Here $r = 3$ (check/parity count), $n = 2^r - 1 = 7$ (code length), $k = n - r = 4$ (data width), and the canonical base $b = n + r = 10$. We use $\{n, k, r\}$ for the Hamming-family scaffold; in the length-7 case this matches the familiar $[7, 4, 3]$ notation because the minimum distance is also 3. Thus

$$142857 = r^3(n + k)(2n - 1)(2^k + nr). \tag{5}$$

This factorization is not a numerical coincidence imposed post-hoc. It arises from the cyclotomic decomposition of $b^{n-1} - 1$ evaluated at $b = n + r$, and we prove the alignment is unique to $r = 3$.

2 Cyclotomic Decomposition

Since $n-1=6$, we have

$$b^6-1=\Phi_1(b)\,\Phi_2(b)\,\Phi_3(b)\,\Phi_6(b)$$

where Φ_d denotes the d-th cyclotomic polynomial. At $b=10$:

$$\Phi_1(10)=10-1=9=3^2, \tag{6}$$
$$\Phi_2(10)=10+1=11, \tag{7}$$
$$\Phi_3(10)=10^2+10+1=111=3\cdot 37, \tag{8}$$
$$\Phi_6(10)=10^2-10+1=91=7\cdot 13. \tag{9}$$

Therefore $10^6-1=9\cdot 11\cdot 111\cdot 91=999999$, and dividing by $n=7$:

$$\frac{10^6-1}{7}=3^3\cdot 11\cdot 13\cdot 37=142857.$$

Matching cyclotomic factors to Hamming parameters:

Cyclotomic factor	Value	Hamming expression
$\Phi_1(10)=9$	3^2	r^2
$\Phi_2(10)=11$	11	$n+k$
$\Phi_3(10)=111$	$3\cdot 37$	$r(2^k+nr)$
$\Phi_6(10)=91$	$7\cdot 13$	$n(2n-1)$

After division by $n=7$ (absorbing one factor from Φ_6) and collecting the factors of 3 (r^2 from Φ_1 and r from Φ_3), we obtain identity (5).

3 Uniqueness

Definition 1 (Self-describing)**.** A Fermat quotient $Q=(b^{n-1}-1)/n$ is *self-describing* relative to a Hamming-family scaffold when its cyclotomic factor-blocks are exhausted by scaffold expressions in $\{n,k,r\}$.

Definition 2. For $r\geq 2$, define the *Hamming-family parameters* $n=2^r-1$, $k=n-r$, and the *canonical base* $b=n+r$.

We require n to be prime (so that $\gcd(b,n)=1$ and $1/n$ has a well-defined repetend in base b). Define $Q=(b^{n-1}-1)/n$, the Fermat quotient. When $\mathrm{ord}_n(b)=n-1$, Q is the full-period cyclic repetend; otherwise it is the Fermat quotient over the full exponent window.

Theorem 1. *Among Hamming-family parameters with $n=2^r-1$ prime and $r\geq 2$, the prime-power factor-blocks of Q decompose into expressions of $\{n,k,r\}$ only at $r=3$.*

Proof. We establish three conditions on the cyclotomic factors. The first alone suffices for uniqueness; the second and third provide independent structural confirmation.

Condition 1 (uniqueness spine). We require $\Phi_2(b)=b+1=n+k$. Since $b=n+r$ and $k=n-r$:

$$n+r+1=n+(n-r)=2n-r,$$

which simplifies to $n = 2r + 1$, equivalently

$$2^{r-1} = r + 1. \tag{10}$$

At $r = 3$: $2^2 = 4 = 4$. ✓

For $r \geq 4$: $f(r) = 2^{r-1} - (r+1)$ satisfies $f(4) = 3 > 0$, and $f(r+1) - f(r) = 2^{r-1} - 1 > 0$ for $r \geq 2$, so f is strictly increasing for $r \geq 3$. Thus $r = 3$ is the unique solution.

Condition 2 (independent confirmation). We require $n \mid \Phi_6(b)$ with $\Phi_6(b)/n = 2n - 1$. Since $\Phi_6(b) = b^2 - b + 1 = (n+r)^2 - (n+r) + 1$, expanding and setting equal to $n(2n-1)$ yields

$$n(n - 2r) = r^2 - r + 1. \tag{11}$$

At $r = 1$: $1 \cdot (1 - 2) = -1 \neq 1$. At $r = 2$: $3 \cdot (3 - 4) = -3 \neq 3$. At $r = 3$: $7 \cdot (7 - 6) = 7 = 7$. ✓

For $r \geq 4$: $n = 2^r - 1 \geq 3r$, so $n - 2r \geq r$, giving $n(n - 2r) \geq 3r \cdot r = 3r^2 > r^2 - r + 1$. No further solutions.

Condition 3 (for prime r). We require $r \mid \Phi_3(b)$ where $\Phi_3(b) = b^2 + b + 1$. Since $b \equiv n \pmod{r}$ and $n = 2^r - 1$:

$$\Phi_3(b) \equiv n^2 + n + 1 = 2^{2r} - 2^r + 1 \pmod{r}.$$

By Fermat's little theorem (for prime r): $2^r \equiv 2 \pmod{r}$, so $2^{2r} \equiv 4 \pmod{r}$, giving

$$\Phi_3(b) \equiv 4 - 2 + 1 = 3 \pmod{r}.$$

Thus $r \mid \Phi_3(b)$ requires $r \mid 3$, and for prime $r \geq 2$ this gives $r = 3$.

Remark 1. Condition 3 uses Fermat's little theorem and therefore applies only to prime r. In the theorem's domain, $n = 2^r - 1$ prime implies r prime (contrapositive: r composite implies $2^r - 1$ composite), so the Fermat step is licensed. Composite solutions to the divisibility condition alone exist (e.g., $r = 57$), but fall outside the hypothesis. In any case, Condition 1 already establishes uniqueness unconditionally.

The unique simultaneous solution is $r = 3$, giving the $[7, 4, 3]$ Hamming code in base $b = 10 = 7 + 3$. □

Corollary 2. *Base* $10 = n + r$ *is the canonical-base candidate for the* $\{7, 4, 3\}$ *scaffold. More generally, each Hamming-family code has canonical-base candidate* $b = 2^r - 1 + r$*, but only at* $r = 3$ *does the resulting Fermat quotient exhibit the self-describing factorization.*

4 Supplementary Observations

4.1 Half-period split (Midy-type)

The repetend splits as $142 \,|\, 857$ with

$$142 + 857 = 999 = 10^3 - 1$$

(a consequence of Midy's theorem for even-period primes) and

$$857 - 142 = 715 = 5 \cdot 11 \cdot 13.$$

The difference factors as $5 \cdot (n + k) \cdot (2n - 1)$, coupling the base-10 cofactor (5, since $10 = 2 \times 5$) with two scaffold primes.

The split has a compact derivation. Let $Q = (10^3 + 1)/7 = 143 = 11 \times 13$. Then $142 = Q - 1$ and $857 = 1000 - Q$, so:

$$857 - 142 = 1001 - 2Q = 5Q = 5 \times 11 \times 13.$$

The same residual 5 appears in both the split difference and the half-cycle correction below.

4.2 Half-cycle correction

$$\frac{1}{7} - \frac{142}{999} = \frac{5}{6993} = \frac{5}{7 \times 999}.$$

The correction from half-cycle repetition to the exact value is the base cofactor divided by the code length times the nines complement.

4.3 Digit invariants

The digit set $\{1, 2, 4, 5, 7, 8\}$, invariant under cyclic rotation, has

$$\text{sum} = 27 = 3^3 = r^3, \qquad \text{product} = 2240 = 2^6 \cdot 5 \cdot 7.$$

5 Computational Null Test

We verified that no other full-reptend prime $p \leq 47$ in base 10 produces a cyclic number whose complete factorization decomposes into expressions of its own structural parameters. Additionally, $p = 31$ (the next Mersenne-prime Hamming length, $31 = 2^5 - 1$) is not full-reptend in base 10: its period is 15, not 30. See the accompanying verification code.

References

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